

# The Three Forms of Linear Equations and Finding the Slopes of Lines (Homework)

① Identify the form of each linear equation below by writing "S" for Standard, "SI" for slope-intercept, or "PS" for point-slope.

I)  $3x - 2 = y$

Answers

SI

II)  $-7x - 8y = -10$

S

III)  $3 = x - y$

S

IV)  $\frac{4}{3}(x+4) = y-11$

PS

V)  $y+1 = 2(x+1)$

PS

VI)  $3.5x = y$

SI

VII)  $\frac{9x}{11} + \frac{y}{3} = -1$

S

VIII)  $-(x-6) = y+5$

PS

IX)  $y = \frac{x}{4}$

SI

X)  $y = -\frac{2}{3}x$

SI

② Identify the slope and y-intercept of each of the following equations.

I)  $y = 2x + 5$

$m=2$   $b=5$

II)  $-3x = y$

$m=-3$   $b=0$

III)  $y = \frac{1}{5}x - 1.3$

$m=\frac{1}{5}$   $b=-1.3$

IV)  $-\frac{x}{3} - \frac{4}{7} = y$

$m=-\frac{1}{3}$   $b=-\frac{4}{7}$

V)  $y = -\frac{4}{5}x + 101$

$m=-\frac{4}{5}$   $b=101$

③ Write linear equations with the following slopes and y-intercepts.

I)  $m=-6$ ,  $b=9$

$y=-6x+9$

II)  $m=-\frac{3}{2}$ ,  $b=0$

$y=-\frac{3}{2}x$

III)  $m=0.45$ ,  $b=\frac{1}{2}$

$y=0.45x+\frac{1}{2}$

IV)  $m=-1$ ,  $b=0.1$

$y=-x+0.1$

V)  $m=7,000$ ,  $b=7.7$

$y=7000x+7.7$

④ Find the slope of each of the following linear equations and a point on the line produced by graphing the equation.

I)  $4(x+5) = y+11$

$m=4$ ,  $(-5,-11)$

II)  $y-1 = -2(x+7)$

$m=-2$ ,  $(-7,1)$

III)  $\frac{2}{9}(x-4) = y-5$

$m=\frac{2}{9}$ ,  $(4,5)$

IV)  $-(x+14) = y+5.99$

$m=-1$ ,  $(-14,-5.99)$

V)  $-5(x-6) = y$

$m=-5$ ,  $(6,0)$



⑤ In each problem below, write an equation of the line with the given slope and point.

I)  $m = -0.4, (3, 9)$

$y - 9 = -0.4(x - 3)$

II)  $m = \frac{1}{2}, (-1, 2)$

$y - 2 = \frac{1}{2}(x + 1)$

III)  $m = 26, (-3, -10)$

$y + 10 = 26(x + 3)$

IV)  $m = -1, (7.4, -4.4)$

$y + 4.4 = -(x - 7.4)$

V)  $m = 6, (3, 0)$

$y = 6(x - 3)$

A) Convert the following linear equations from standard form to slope-intercept form.

B) Then find the slope and y-intercept of the lines produced by graphing the equations.

⑥  $7x + y = 22$

A)  $y = -7x + 22$  B)  $m = -7, b = 22$

⑦  $-15x - 3y = 2$

A)  $y = -5x - \frac{2}{3}$  B)  $m = -5, b = -\frac{2}{3}$

⑧  $2x + 6y = -1$

A)  $y = -\frac{1}{3}x - \frac{1}{6}$  B)  $m = -\frac{1}{3}, b = -\frac{1}{6}$

⑨  $x + 2y = 3$

A)  $y = -\frac{1}{2}x + \frac{3}{2}$  B)  $m = -\frac{1}{2}, b = \frac{3}{2}$

⑩  $-5x + 2y = 9$

A)  $y = \frac{5}{2}x + \frac{9}{2}$  B)  $m = \frac{5}{2}, b = \frac{9}{2}$

A) Convert the following equations from point-slope form to slope-intercept form. B) Then find the slope and y-intercept.

⑪  $-3(x - 2) = y + 1$

A)  $y = -3x + 5$  B)  $m = -3, b = 5$

⑫  $\frac{1}{2}(x + 4) = y - 3$  A)  $y = \frac{1}{2}x + 5$  B)  $m = \frac{1}{2}, b = 5$

⑬  $-7(x - 6) = y - 8$  A)  $y = -7x + 50$  B)  $m = -7, b = 50$

⑭  $-(x + 3) = y + 7$  A)  $y = -x - 10$  B)  $m = -1, b = -10$

⑮  $11(x + 1) = y - 1$  A)  $y = 11x + 12$  B)  $m = 11, b = 12$

In each problem below, use the slope equation to find the slope of the line that passes through the two given points.

⑯  $(10, 9)$  &  $(8, 11)$

$m = -1$

⑰  $(1, -7)$  &  $(-2, 0)$

$m = -\frac{7}{3}$

⑱  $(-4, -3)$  &  $(-5, -6)$

$m = 3$

⑲  $(2, 1)$  &  $(12, -2)$

$m = -\frac{3}{10}$

⑳  $(-10, 10)$  &  $(6, 4)$

$m = -\frac{3}{8}$

A) In each problem below, use the slope equation to find the slope of the line that contains the two given points. B) Then write a point-slope form of the equation of the line containing the two given points.

C) Then convert the equations from point-slope form to slope-intercept form.

㉑  $(4, 7)$  &  $(2, -3)$

A)  $m = 5$  B)  $y - 7 = 5(x - 4)$

or  $y + 3 = 5(x - 2)$

C)  $y = 5x - 13$



②②  $(-1, -2)$  &  $(-6, 2)$

A)  $m = -\frac{4}{5}$

B)  $y + 2 = -\frac{4}{5}(x + 1)$

or  $y - 2 = -\frac{4}{5}(x + 6)$

C)  $y = -\frac{4}{5}x - \frac{14}{5}$

②③  $(0, 0)$  &  $(3, 1)$

A)  $m = \frac{1}{3}$

B)  $y - 0 = \frac{1}{3}(x - 0)$

or  $y - 1 = \frac{1}{3}(x - 3)$

C)  $y = \frac{1}{3}x$

②④  $(-6, -1)$  &  $(-2, -5)$

A)  $m = -1$

B)  $y + 1 = -(x + 6)$

or  $y + 5 = -(x + 2)$

C)  $y = -x - 7$

Use the slope-intercept equation and the given information to find the slope-intercept form of the equation of the corresponding line.

②⑤  $(2, 3)$ ,  $b = 1$

$y = x + 1$

②⑥  $(-1, 5)$ ,  $m = -2$

$y = -2x + 3$

②⑦  $(-6, -8)$ ,  $b = -\frac{3}{4}$

$y = \frac{29}{24}x - \frac{3}{4}$

②⑧  $(1, -4)$ ,  $m = 6$

$y = 6x - 10$

②⑨  $(10, 3)$ ,  $b = 0$

$y = \frac{3}{10}x$